

Three-observable decoder for reset masks

Exact matched-parameter result for $M>0$, $Pe\geq 0$, $\rho>0$; G is the centered spatial growth class.

Decoder: $\Sigma = (G, U_{xx}, \text{Tr } S)$ recovers P, Θ , then V .
Localized/diffusive $\rightarrow$ orientation-reset status $\rightarrow$ velocity-reset status.

map	G	U_{xx}	Tr S
P	localized	$1/2$	A
PV	localized	$1/2$	B
P Θ	localized	$(\rho+2)/(\rho+4)$	C
PV Θ	localized	$(\rho+2)/(\rho+4)$	B
V	diffusive	$1/2$	B
Θ	diffusive	$(\rho+2)/(\rho+4)$	C
V Θ	diffusive	$(\rho+2)/(\rho+4)$	B

Exact strict gaps
 $(\rho+2)/(\rho+4) - 1/2 = \rho/[2(\rho+4)] > 0$; $A-B > 0$; $C-B > 0$.
The decoder remains injective at the passive point $Pe=0$ and on the resonance $M\rho=1$.